ENGR 225 Homework 5 Questions

Assoc. Prof. Clayton Byers

Assigned: 16 Oct 2026

Due: 23 Oct 2026 (11:59pm on Moodle)

Important notes before you begin:

- Make sure you are drawing your Free Body Diagrams (FBDs) whenever analyzing system equilibrium. That includes single points as well as larger 2- and 3-D systems.
- Problem 1 will have 6 equations for 6 unknowns. That means you either have to methodically plug equations into each other, or use linear algebra. You are not expected to know linear algebra, but if you do it is a powerful tool.
- Problem 8 is technically doable once you've learned about the method of sections and joints, but due to me being out of town we won't see an example in class until after this is due. Try to tackle this problem ahead of time, but don't worry if you struggle with it, I won't grade this problem.

Problem 1

The three dimensional rod ABC is supported by smooth journal bearings at A, B, and C. Determine the magnitude of F_2 which will cause the reaction C_y at the bearing C to be zero. Assume the bearings are properly aligned so they exert forces only, no moments. Set $F_1 = 300\,lb$. (**Ans:** $F_2 = 674\,lb$)

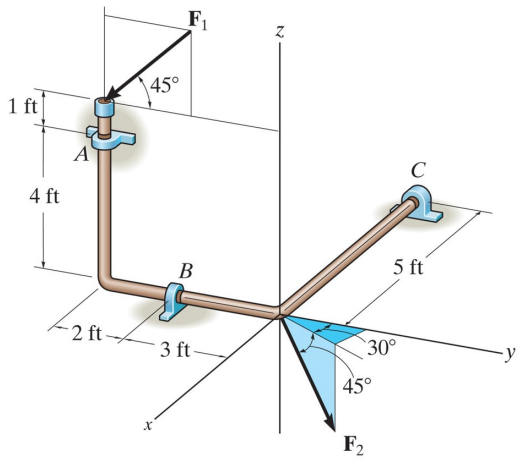


Figure for problem 1.

Problem 2

Member AB is supported by a cable BC and a square rod at A which fits loosely through the square hole a the end joint as shown. Determine the component reactions at A and the tension in the cable required to hold the $800\,lb$ cylinder in equilibrium. (**Ans:** $A_y = 0$, $A_z = 800\,lb$, $M_x = 4800\,lb \cdot ft$, $M_y = M_z = 0$, $T_{BC} = 0$.)

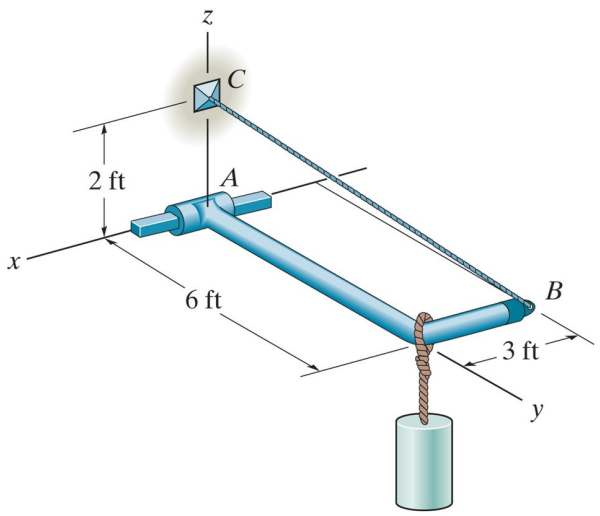


Figure for problem 2.

Problem 3

Determine the force in each member of the truss and indicate whether they are in tension or compression. Set $\theta = 30°$. (**Ans:** $F_{AB} = 2.36\,kN\,(T)$, $F_{AD} = 1.46\,kN\,(C)$, $F_{BC} = 2.36\,kN\,(T)$, $F_{BD} = 4.00\,kN\,(T)$, and $F_{CD} = 5.21\,kN\,(C)$)

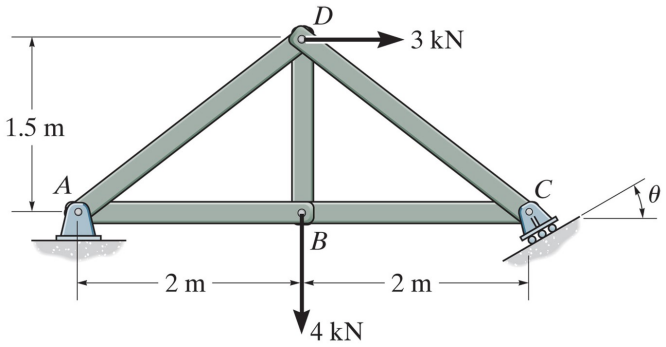


Figure for problem 3.

Problem 4

Members AB and BC can each support a maximum compressive force of $800\,lb$, while AD, DC, and BD can each support a maximum tensile force of $2000\,lb$. If $a = 6\,ft$, determine the greatest load P that can be applied to the truss. (**Ans:** $P = 848\,lb$)

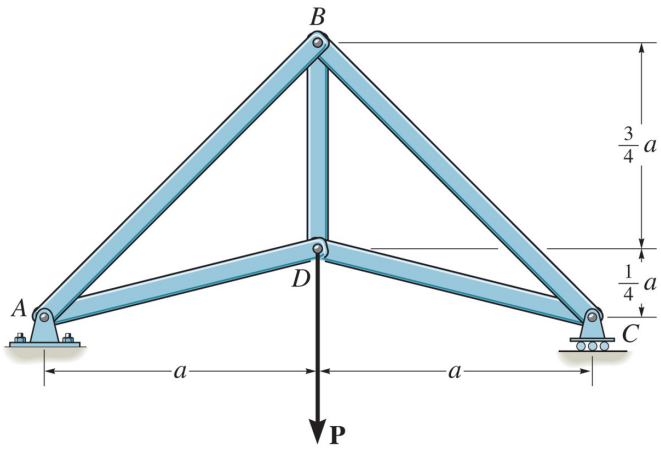


Figure for problem 4.

Problem 5

The maximum allowable tensile force in any of the members of the truss is $5\,kN$ and the maximum allowable compressive force is $3\,kN$. Determine the maximum load P that can be applied to the truss. Set $d = 2\,m$. (**Ans:** $P = 4.16\,kN$)

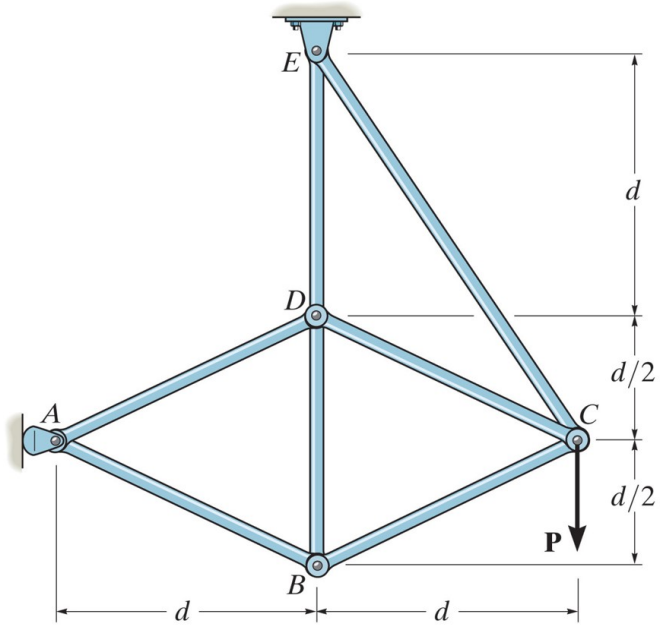


Figure for problem 5.

Problem 6

Determine the force in members EF, CF, and BC and state if they are in tension or compression. (**Ans:** $F_{BC} = 5.33\,kN\,(C)$, $F_{CF} = 4.00\,kN\,(T)$, and $F_{EF} = 5.33\,kN\,(T)$)

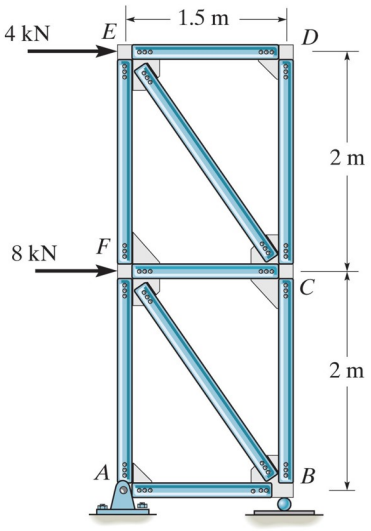


Figure for problem 6.

Problem 7

Determine the force in members BF, BG, and AB and state if they are in tension or compression. (**Ans:** $F_{AB} = 45.0\,kN\,(T)$, $F_{BF} = 0\,kN$, and $F_{BG} = 35.4\,kN\,(C)$)

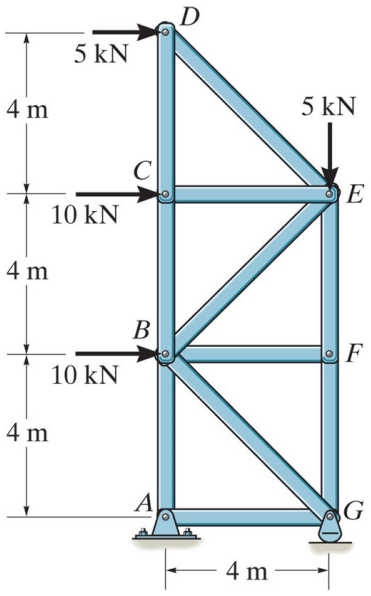


Figure for problem 7.

Problem 8

Determine the force in each member of the space truss and state if they are in tension or compression. The truss is supported by ball and socket joints at A, B, C, and D. (**Ans:** $F_{AE} = 6.71\,kN\,(C)$, $F_{BE} = 0$, $F_{CG} = 4.47\,kN\,(T)$, $F_{DE} = 9.00\,kN\,(T)$, $F_{DG} = 4.47\,kN\,(C)$, and $F_{EG} = 6.00\,kN\,(C)$)

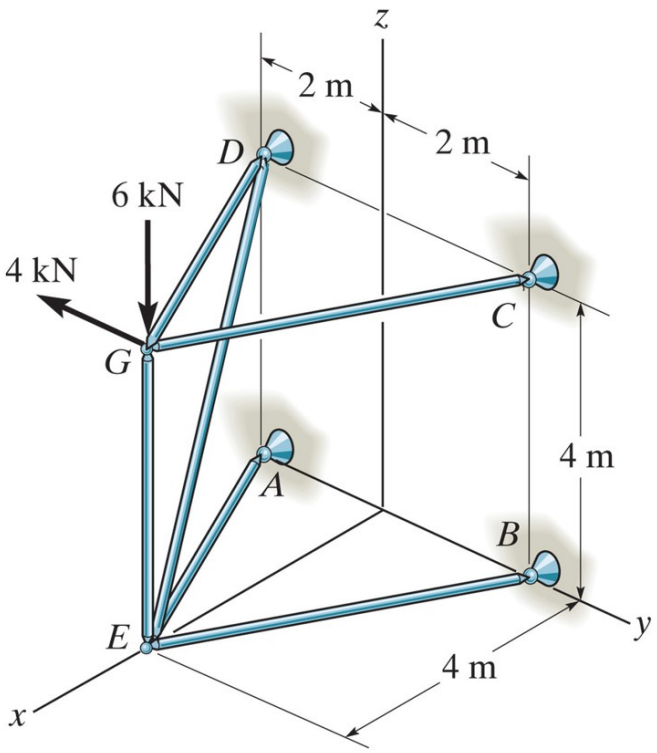


Figure for problem 8.